Title of the talk which can span over more lines

First Author[1], Second Author[1,2] and Last Author[2]

[1]Department of Materials Science, Montanuniversität Leoben, Austria
[2]Name of the second institution, City, Country

name.surname@unileoben.ac.at

Name of the Conference
1 January 2026

Outline

1. Overview
2. Basics
3. Lists and overlays
4. Figures
5. More tools

Class options

Footer	`defaultFootings`*, `fullFootings`, `lectureFootings`, `emptyFootings` (see below)
Slide numbers	`slideSingleNumber`*, `slideTotalNumber` (current/total)
Sidebar	`showSidebar`*, `hideSidebar`; `showothersubsections`*, `hideothersubsections`
Language	`DElocalization`*, `ENlocalization` (logo, claim, texts); `DElogo`, `ENlogo`, `DEslogan`, `ENslogan` (logo or claim only)
Title slide	`lightTitle`*, `darkTitle`
Text font	`classicFont`* (Latin Modern Sans), `brandFont` (Host Grotesk)
Results	`showResults`*, `hideResults` (`\showResults`)
beamer	all other options are passed to beamer, e.g. `trans`, `handout`; the aspect ratio is always 16:9

* default

`defaultFootings`



`fullFootings`



`lectureFootings`



`emptyFootings`



with `fullFootings` in lecture mode, the conference is replaced by the current section/subsection, e.g. *1.2 Subsection*

Commands

Title slide	`\title[<short>]`, `\author[<short>]` with `\aff{<n>}`, `\institute[<unit for footer>]`, `\email`, `\date`, `\conference` (talk) or `\lecture` (lecture mode), `\additionsToTitlePage`
Settings	`\setFrameLogo`, `\setTitleLogo`, `\setSlogan`, `\setThanksText`, `\setAgendaTitle`, `\setDarkBackground`, `\website`, `\setQRcode`
Special slides	`\agendaframe`, `\personframe`, `\finalframe` (between frames)
Lectures	`\lectureBeamer` with `\includeonlylecture` (beamer's original `\lecture`)
Content	`\alert`, `\Alert`, `\example`, `\funneldisplay`, `\showResults`, colours `digital_…`
Between frames	`\literature`, `\lastslide`
Placing	`overdraw` environment, `\framegrid`

All of them are described in `README.md` and in the backup slides at the end (`demo_documentation.tex`); the following slides demonstrate them.

Text and highlighting

- important content is highlighted with `\alert{…}`
- less important content is highlighted with `\Alert{…}`
- The MUL display font (Funnel Display) is switched on with `\funneldisplay`; it is used for the frame titles
- the text is set in Latin Modern Sans; class option `brandFont` switches to Host Grotesk, the MUL text font
- mathematics works as usual: $a^2 + b^2 = c^2$, $\sin\alpha = \frac{b}{c}$

Blocks and columns

Blocks and columns

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

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Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

An example block

For examples.

Blocks and columns

A block

Standard beamer `block` in the MUL colours.

An alert block

For warnings and important statements.

An example block

For examples.

A block without title.

Colours

The brand colours are defined with the names of the MUL style guide (also used by the poster template):

Use them as usual, e.g. `\textcolor{digital_Turquoise}{…}`.

Colours

The brand colours are defined with the names of the MUL style guide (also used by the poster template):

primary

digital_Turquoise

digital_GraphiteBlack

digital_LightGreen

digital_TechGreen

digital_White

Use them as usual, e.g. `\textcolor{digital_Turquoise}{…}`.

Colours

The brand colours are defined with the names of the MUL style guide (also used by the poster template):

primary

digital_Turquoise

digital_GraphiteBlack

digital_LightGreen

digital_TechGreen

digital_White

combination (with a primary colour)

digital_BTLight

digital_BTDark

digital_TGLight

digital_TGDark

digital_GWLight

digital_GWDark

Use them as usual, e.g. `\textcolor{digital_Turquoise}{…}`.

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digital_GWDark

secondary (study areas only)

digital_Violet

digital_Orange

digital_Blue

Use them as usual, e.g. `\textcolor{digital_Turquoise}{…}`.

Lists

By default, list items are uncovered one by one (<+-> is the class default):

- first level items have squares

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Lists

By default, list items are uncovered one by one (<+-> is the class default):

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  - lower levels have dashes

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- first level items have squares
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- *Example:* examples are marked with `\example` instead of `\item`

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Lists

By default, list items are uncovered one by one (`<+->` is the class default):

- first level items have squares
  - lower levels have dashes
- *Example:* examples are marked with `\example` instead of `\item`
- last item

Use `\begin{itemize}[<1->]` to show all items at once:

- all
- at
- once

Overlays

- `\only<2>{…}`: content exists on slide 2 only (takes no space otherwise)
- `\uncover<3->{…}`: space for the content is always reserved, the content is shown from slide 3 on

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This text is shown with `\only<2>`.

Overlays

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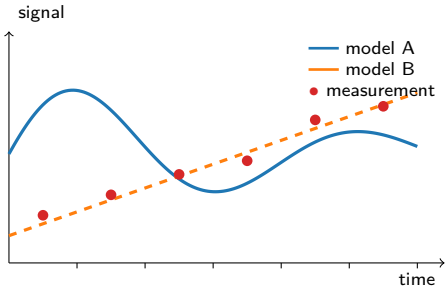
This text is shown with `\uncover<3->`.

Figures and literature

- figures are kept in the folder `figs`, graphics (logos, background) in `graphics`
- images are included with `\includegraphics{figs/…}`; these demo figures are TikZ drawings included with `\input`
- `\literature{…}` (outside a frame) prints references above the footer of this and all following slides; clear it with `\literature{}`

A. Author, B. Author, and C. Author, *J. Example Sci.* **1**, 1 (2026)
D. Author and E. Author, *Example Lett.* **2**, 100 (2025)

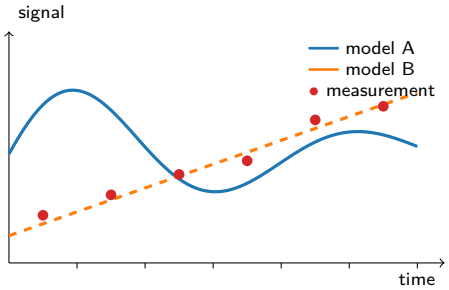
Figures and literature

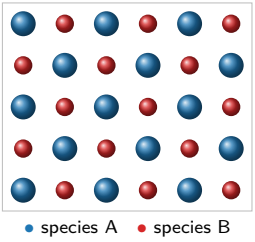


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Figures and literature





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D. Author and E. Author, *Example Lett.* **2**, 100 (2025)

Placing objects with `overdraw`

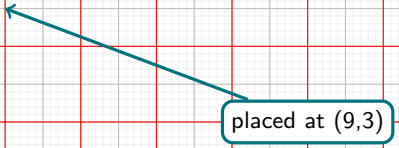
- the `overdraw` environment is a TikZ picture with absolute coordinates: one unit is 1/16 of the slide width, the origin is the bottom-left corner of the slide (right of the sidebar, if shown)
- `\framegrid` inside `overdraw` draws the coordinate grid for finding positions – remove it in the final version
- steps: grid (slide 2), grid with label and arrow placed with `overdraw` (slide 3), final result (slide 4)

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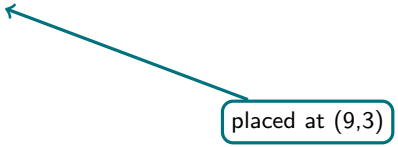
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Placing objects with `overdraw`

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- `\framegrid` inside `overdraw` draws the coordinate grid for finding positions – remove it in the final version
- steps: grid (slide 2), grid with label and arrow placed with `overdraw` (slide 3), final result (slide 4)



Hidden results

- `\showResults{…}` shows its content with the class option `showResults` (default), and leaves an empty space of the same size with `hideResults`
- independent of the beamer mode: in the presentation (default) as well as with the beamer class options `trans` or `handout` (one page per frame, without overlays, e.g. for printing), only `showResults`/`hideResults` decide
- handouts of exercises without results: `\documentclass[trans, hideResults]{mul-beamer}`

Exercise

Solve $x^2 - 5x + 6 = 0$.

Result: $x_1 = 2, \quad x_2 = 3$

Frame logo

- the logo in the top-right corner can be changed with `\setFrameLogo{graphics/…}`
- here: the logo of the Department of Materials Science

Agenda slide

- `\agendaframe` (between frames) shows the table of contents (sections only) in the *Agenda* layout of the MUL PowerPoint master, see the second slide
- `\agendaframe[<title>]` replaces the title (default: *Outline*); `\setAgendaTitle{<title>}` (in the preamble) changes the default

Person slide

- `\personframe[<photo>]{<name>}{<role>}{<text>}` (between frames) introduces a person, as in the *Zur Person* layout of the MUL PowerPoint master
- the photo fills the left part of the slide (cropped as needed); without `<photo>`, a placeholder is shown (next slide)
- `<text>` can contain lists, e.g. a short CV

Name Surname

Function, Department of Materials Science

- since 2020: current position
- 2015–2020: previous position
- research interests: topic A, topic B

Closing slide

- `\finalframe` (between frames) adds a closing slide in the dark style of the MUL slides – see the end of `talk.tex`: "Thank you for your attention!" (localised), for talks the title and the authors, the e-mail address (`\email`), the website (`\website`, default unileoben.ac.at), an optional QR code (`\setQRcode`), and the claim and author line in the footer
- `\finalframe[…]` replaces the text, e.g. `\finalframe[Questions?]`
- `\setDarkBackground{graphics/background_dark.jpg}` (in the preamble) uses the background of the official PowerPoint master instead of the CMS variant
- the class option `darkTitle` gives a title slide in the same dark style

Thank you for your attention!

unileoben.ac.at

Title of the talk which can span over more lines

First Author, Second Author and Last Author

name.surname@unileoben.ac.at
unileoben.ac.at

Backup slide

- slides after `\lastslide` are not counted in the total number of slides

Backup slide

- slides after `\lastslide` are not counted in the total number of slides
- useful for backup slides

Documentation: class options (1/2)

Footer (* default)	
`defaultFootings`*	slogan, author | organisational unit (`\institute[...]`), slide number
`fullFootings`	slogan, author: short title, conference (lecture mode: current section/subsection), date, slide number
`lectureFootings`	slogan, lecture name (`\lecture{...}`), slide number
`emptyFootings`	slide number only
Slide numbers	
`slideSingleNumber`*	current slide number
`slideTotalNumber`	current/total; the total counts up to `\lastslide`
Sidebar	
`showSidebar`*, `hideSidebar`	sidebar with the table of contents, or none
`showothersubsections`*, `hideothersubsections`	subsections of all sections, or of the current section only, in the sidebar
Title slide	
`lightTitle`*	light background, turquoise title, grey claim
`darkTitle`	dark background (`\setDarkBackground`), white logo and text, claim in Light-Green

Documentation: class options (2/2)

Language (* default)	
`DElocalization`*, `ENlocalization`	German or English logo, claim, and texts (closing slide, agenda title)
`DElogo`, `ENlogo`	title logo only (overrides the localization)
`DEslogan`, `ENslogan`	claim only (overrides the localization)
Text font	
`classicFont`*	Latin Modern Sans for text, and letters, digits, operator names in formulas
`brandFont`	Host Grotesk (MUL brand font, `fonts/Host_Grotesk/`); fallbacks: installed Host Grotesk, Calibri, Carlito, Latin Modern Sans
Results	
`showResults`*, `hideResults`	content of `\showResults{...}` shown, or replaced by empty space; independent of the beamer mode
Other options e.g. `trans`, `handout`	passed to beamer (one page per frame, without overlays); the font size (10 pt), top-aligned frames, and the aspect ratio 16:9 are fixed

Example

```
\documentclass[ENlocalization, DElogo, DEslogan, hideSidebar]{mul-beamer}
```

Documentation: title slide

- `\title[<short title>]{<title>}`, `\author[<short authors>]{<authors>}`, `\date{<date>}` – short versions for the footer (`fullFootings`, closing slide)
- `\aff{<n>}` – affiliation number (superscript) in `\author` and `\institute`; omitted in the PDF metadata and on the closing slide
- `\institute[<unit for the footer>]{<affiliations>}` – give `[...]` for multi-line affiliations (otherwise the full text appears in `defaultFootings`)
- `\email{<address>}` – on the title slide and the closing slide (as a link)
- `\conference{<name>}` – talk: on the title slide and in `fullFootings`
- `\lecture{<name>}` – lecture mode: table of contents at each section, name in `lectureFootings`, current section in `fullFootings`; the name is not shown on the title slide (put it into `\title`)
- `\additionsToTitlePage{<TikZ nodes>}` – e.g. partner logos:

Example

```
\additionsToTitlePage{\node[anchor=north west, shift={(0.04\paperheight,
  -0.04\paperheight)}] at (current page.north west)
  {\includegraphics[width=0.084\paperwidth]{graphics/logo-partner.pdf}};}
```

Documentation: settings (preamble)

- `\setFrameLogo{<file>}` – logo in the frame titles (in a group `{...}` around frames: these frames only)
- `\setTitleLogo{<logo>}{<white logo>}` – logo of the title slide and of the dark slides
- `\setSlogan{<text>}` – claim on the title slide, in the footer, and on the closing slide; `\setSlogan{}` removes it
- `\setThanksText{<text>}` – default text of `\finalframe`
- `\setAgendaTitle{<title>}` – default title of `\agendaframe` (Inhalt/Outline)
- `\setDarkBackground{<file>}` – background of the dark slides: `graphics/background_dark_cms.jpg` (default) or `graphics/background_dark.jpg` (PowerPoint master)
- `\website[<url>]{<text>}` – website on the closing slide (default unileoben.ac.at; `<url>` defaults to `https://<text>`; `\website{}` removes it)
- `\setQRcode[<caption>]{<image file>}` – QR code on the closing slide, e.g. linking to the group website or a paper; create static QR codes e.g. at `https://www.tu-chemnitz.de/urz/apps/qrcode/index.html.en` (PNG, or SVG converted by `make`)

Documentation: special slides

- `\agendaframe[<title>]` – table of contents (sections only), *Agenda* layout of the PowerPoint master
- `\personframe[<photo>]{<name>}{<role>}{<text>}` – *Zur Person* layout: photo on the left (placeholder without `<photo>`), name, role, and text (may contain lists)
- `\finalframe[<text>]` – closing slide: text, title and authors (talks), e-mail, website, QR code, claim and author line

Example

```
\agendaframe
\personframe[figs/photo.jpg]{Name Surname}{Function}{Short CV}
\finalframe[Questions?]
```

Documentation: content of frames

- `\alert{...}` – important content (red); `\Alert{...}` – less important content (turquoise)
- `\example` – instead of `\item`: item marked with a stop sign and *Example:*
- `\funneldisplay` – switches to the display font Funnel Display (used for titles and the claim)
- `\showResults{...}` – content shown with `showResults`, empty space with `hideResults`
- lists are uncovered item by item by default (`<+->`); `\begin{itemize}[<1->]` shows all items at once
- colours of the style guide: `digital_Turquoise`, `digital_GraphiteBlack`, `digital_LightGreen`, `digital_TechGreen`, `digital_White`; combination colours `digital_BTLight`, …, `digital_GWDark`; secondary colours `digital_Violet`, `digital_Orange`, `digital_Blue` (study areas only); for figures `plotblue`, `plotgreen`, `plotorange`, `plotred`, `plotpurple`, `plotbrown`

Documentation: further commands

- `\literature{<references>}` – references above the footer of all following slides; `\literature{}` clears them
- `\lastslide` – the total number of slides (`slideTotalNumber`) counts only up to here (backup slides)
- `\lectureBeamer[<short>]{<title>}{<label>}` with `\includeonlylecture{<label>}` – beamer's original `\lecture`: several lectures in one file, only the selected one is compiled
- `overdraw` environment – TikZ picture with absolute coordinates (unit 1/16 of the slide width, origin bottom left, right of the sidebar); `\framegrid` inside it shows the grid:

Example

```
\begin{overdraw}
  \framegrid   % remove in the final version
  \node[draw=digital_Turquoise] (a) at (9,3) {label};
  \draw[->, digital_Turquoise] (a) -- (5,4.5);
\end{overdraw}
```